D.A.V. PUBLIC SCHOOL, SECTOR IV, B. S. CITY

CLASS XI — MATHEMATICS REVISION SHEET (SET 2)

QUESTION BANK VARIANT B **TOPIC-WISE STRUCTURED COMPILATION**

GENERAL INSTRUCTIONS:

1. This question bank contains topic-wise numerical variations of standard board pattern problems to reinforce conceptual mastery in Class XI Mathematics.
2. Solve all problems sequentially to build fluency across Straight Lines, Conics, and Three-Dimensional Geometry.

UNIT 1: STRAIGHT LINES

Q1. The slope-intercept form of a straight line equation is expressed as:

(a) $y = mx + c$ (b) $x/a + y/b = 1$

(c) $x \cos \alpha + y \sin \alpha = p$ (d) $ax + by = 0$

Q2. Calculate the perpendicular distance between the given pair of parallel lines:

(i) $5x - 12y + 9 = 0$ and $5x - 12y - 17 = 0$

(ii) $4x + 3y - 11 = 0$ and $4x + 3y + 4 = 0$

(iii) $8x + 15y + 18 = 0$ and $8x + 15y - 16 = 0$

(iv) $k(x + y) + c_1 = 0$ and $k(x + y) + c_2 = 0$

Q3. Prove collinearity of the following sets of points using straight line concepts:

(i) Points (2, 3), (4, 7) and (6, 11)

(ii) Points (−1, 4), (2, 1) and (5, −2)

(iii) Points (p, 0), (0, q) and (2, 2) if $1/p + 1/q = 1/2$

Q4. Image and Reflection Problems:

(i) Find the image of the point (2, 3) with respect to the line $x + 2y = 3$ considering the line as a plane mirror.

(ii) Find the image of the point (4, 1) with respect to the line $3x - y = 4$ acting as a plane mirror.

(iii) Determine the coordinates of the foot of the perpendicular drawn from (2, 5) onto the line $x + 2y - 7 = 0$.

Q5. Perpendicular Distance and Intercept Proofs:

(i) If d is the perpendicular distance from origin to a line making intercepts h and k on the coordinate axes, prove that $1/d^2 = 1/h^2 + 1/k^2$.

(ii) Show that the product of the lengths of perpendiculars drawn from $(\pm\sqrt{(a^2 - b^2)}, 0)$ to $(x/a) \sec \theta + (y/b) \tan \theta = 1$ simplifies systematically.

(iii) Find the equation of the line passing through (4, −2) and perpendicular to the line joining (1, 3) and (5, −1).

UNIT 2: CONIC SECTIONS — CIRCLE & PARABOLA

Q6. Determine the center and radius for each of the given circle equations:

(i) $x^2 + y^2 + 4x - 8y - 5 = 0$

(ii) $x^2 + y^2 - 10x + 6y + 9 = 0$

(iii) $3x^2 + 3y^2 - 6x + 12y - 15 = 0$

Q7. Parabola Parameters and Latus Rectum:

(i) Find the length of the latus rectum of parabola $y^2 = 20x$:

(a) 5 (b) 10

(c) 15 (d) 20

(ii) Find the length of the latus rectum of parabola $x^2 = 16y$:

(a) 4 (b) 8

(c) 12 (d) 16

(iii) Find the focus, axis of symmetry, directrix equation, and latus rectum length of $y^2 = -12x$.

UNIT 2 (CONTD.): CONIC SECTIONS — ELLIPSE & HYPERBOLA

Q8. Standard Equations of Conic Curves:

(i) The standard equation of an ellipse with major axis along y-axis is:

(a) $x^2/a^2 + y^2/b^2 = 1$ $(a > b)$ (b) $x^2/b^2 + y^2/a^2 = 1$ $(a > b)$

(c) $x^2/a^2 - y^2/b^2 = 1$ (d) $y^2 - x^2 = a^2$

Q9. Find the standard equation of the ellipse given the following geometric conditions:

(i) Vertices at $(\pm 8, 0)$ and foci at $(\pm 5, 0)$

(ii) Vertices at $(0, \pm 15)$ and foci at $(0, \pm 9)$

(iii) Ends of major axis at $(\pm 6, 0)$ and ends of minor axis at $(0, \pm 3)$

(iv) Length of major axis 26 and foci located at $(\pm 12, 0)$

Q10. Practical and Applied Conic Problems:

(i) A metallic rod PQ of length 20 cm moves with its ends P and Q sliding on the x-axis and y-axis respectively. A point R(x, y) is chosen on the rod such that PR = 8 cm. Prove that the trajectory of point R is an ellipse.

(ii) A suspension bridge cable is supported by towers 20 m apart. The vertical sag at the center is 4 m and the cable assumes a parabolic shape. Find the height of the cable 5 m away from the center.

(iii) An archway built in the shape of a vertical parabola is 12 m high and 8 m wide at ground level. Calculate its width at a height of 6 m above the base.

UNIT 3: INTRODUCTION TO THREE DIMENSIONAL GEOMETRY

Q11. Fundamental Properties of 3D Space:

(i) The z-coordinate of any point situated on the XY-plane is:

(a) 1 (b) 0

(c) -1 (d) Any arbitrary constant

(ii) Points lying in the Vth octant have coordinates of the sign form:

(a) $(+, +, +)$ (b) $(+, -, +)$

(c) $(+, +, -)$ (d) $(-, +, +)$

(iii) Find the geometric representation of the set of points satisfying $x = 0$ and $y = 0$ simultaneously.

(iv) Identify the octants containing the points $(3, -2, -5)$, $(-4, 2, -1)$, and $(-1, -5, 6)$.

Q12. Distance and Section Formula in 3D Coordinate Geometry:

(i) Calculate the distance between points A(3, −1, 5) and B(−2, 4, 0).

(ii) Show that points P(3, 2, 2), Q(1, 4, 0), R(−1, 1, 2) and S(1, −1, 4) form the vertices of a rectangle.

(iii) Determine the coordinates of the point dividing the segment joining (2, −4, 5) and (5, 6, −1) in the ratio 3 : 2 externally.

(iv) Using section formula, demonstrate that points A(3, 2, −4), B(5, 4, −6) and C(9, 8, −10) are collinear.